

STRATEGY

WHAT SPLITS

IS REPLICATED

COMMUNICATE

TP · tensor	W weights (rows)	activations	partial sums → all-reduce
PP · pipeline	transformer layers	hidden states	hidden states (P2P)
DP · data	data batch	model + optimizer	gradients → all-reduce
EP · expert	MoE experts	attention / weights	routes → all-to-all
CP · context	token sequence	model weights	KV (ring)

The only difference between the strategies is WHICH of these three things is split vs replicated, and what is exchanged. Compose them (DP x TP x PP, or TP+CP in a decode pool + prefill pool). [ILLUSTRATIVE conceptual]